

BOHB: Bayesian Optimisation and Hyperband on YAHPO Gym

Hyperparameter Optimisation for Machine Learning — Assignment 2

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1 Introduction

We extend our Assignment 1 comparison with BOHB [2], which combines the bracket portfolio of Hyperband [5] with model-guided configuration proposal via a Random Forest surrogate [3] and Expected Improvement. We evaluate all seven algorithms across six YAHPO Gym benchmarks [6] and study two surrogate refitting strategies (per-round and per-bracket).

2 Methods

BOHB. BOHB [2] runs the same $s_{\max}+1$ Successive Halving brackets as Hyperband. Before each round, it attempts to fit a Random Forest surrogate (100 trees, `min_samples_leaf` = 3; following [3]) on all observations collected at the highest fidelity with at least $N_{\min} = \max(10, n_{\text{hp}} + 1)$ data points, where n_{hp} is the number of hyperparameters. The next configuration is drawn by maximising Expected Improvement (EI) [7] over a pool of 64 random candidates: $\text{EI}(x) = (\mu - f^*)\Phi(Z) + \sigma\phi(Z)$, $Z = (\mu - f^*)/\sigma$, where μ and σ are the per-candidate mean and standard deviation of tree predictions. A fraction $\rho=1/3$ of configurations is always drawn uniformly at random (Algorithm 2, Falkner et al.), preserving the convergence guarantee when low-fidelity rankings are misleading.

Categorical hyperparameters are *label-encoded* (each mapped to $\text{index}/(|\text{choices}| - 1)$) rather than one-hot encoded. For NB301 this reduces the encoded dimension from 272 to 34, bringing $N_{\min}$ from 273 down to 35 and allowing the surrogate to activate early in the run.

We compare two surrogate refitting strategies: **BOHB-Round** fits the model once at the start of each round; all brackets share the same snapshot. **BOHB-Bracket** refits at the start of each bracket within a round, so later brackets benefit from observations accumulated by earlier ones.

Baselines. Random Search (RS) [1], Grid Search (GS), Successive Halving (SH) [4], Hyperband (HB) [5], and Bayesian Optimisation (BO) with a GP surrogate [7] are as in Assignment 1.

3 Experimental Setup

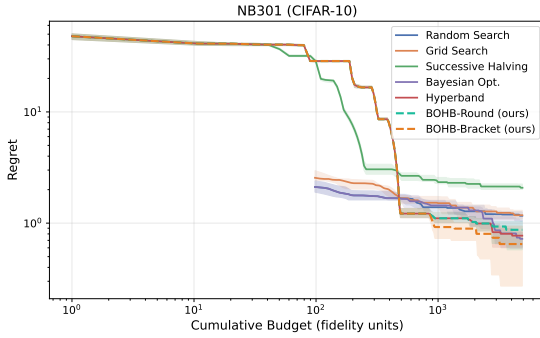
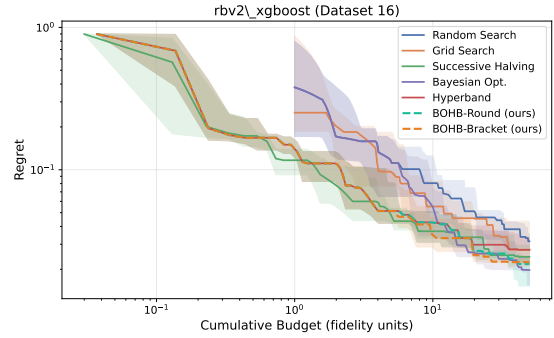
Six YAHPO Gym benchmarks: NB301/CIFAR-10 ($r_{\max}=98$ epochs, val. accuracy), `rbv2_xgboost/16` ($r_{\max}=1$, acc), `rbv2_svm/16`, `rbv2_glmnet/16`, `rbv2_ranger/16` ($r_{\max}=1$, acc), and `LCBench/3945` ($r_{\max}=51$ epochs, val. accuracy). Total budget $B=50 \times r_{\max}$ fidelity units, $\eta=3$, 10 independent seeds per algorithm (420 runs total).

4 Results

NB301. BOHB-Bracket achieves the highest mean (94.24 ± 0.43), narrowly ahead of Hyperband (94.16 ± 0.20), though the difference lies within one standard deviation. BOHB-Round (94.05 ± 0.33) is slightly below Hyperband because the shared model is fitted with fewer observations before the run warms up: the surrogate only activates once $N_{\min}=35$ points are available at a single fidelity, which requires about a full round. BOHB-Bracket mitigates this by refitting before each bracket, so brackets $s>0$ benefit from observations already accumulated. The higher variance of both BOHB variants relative to Hyperband reflects that the RF occasionally proposes a configuration that looks promising under EI but underperforms at full fidelity, a consequence of the limited 35-point warm-up data available on a 34-dimensional space. SH underperforms (92.79 ± 0.17) because its single bracket never reaches $r_{\max}=98$ with $\eta=3$.

Table 1: Final incumbent (mean $\pm$ std, 10 seeds). **Bold** = best per benchmark.

Algorithm	NB301	xgb/16	svm/16	glmnet/16	ranger/16	lcb/3945
Random Search	93.73 $\pm$ 0.17	0.962 $\pm$ 0.005	0.980 $\pm$ 0.002	0.973 $\pm$ 0.001	0.959 $\pm$ 0.003	99.35 $\pm$ 0.13
Grid Search	93.71 $\pm$ 0.18	0.963 $\pm$ 0.010	0.978 $\pm$ 0.003	0.979$\pm$0.005	0.959 $\pm$ 0.004	98.77 $\pm$ 0.72
Succ. Halving	92.79 $\pm$ 0.17	0.968 $\pm$ 0.007	0.976 $\pm$ 0.002	0.975 $\pm$ 0.002	0.964 $\pm$ 0.003	99.48 $\pm$ 0.03
Bayesian Opt.	94.08 $\pm$ 0.32	0.974 $\pm$ 0.009	0.981$\pm$0.002	0.977 $\pm$ 0.001	0.967$\pm$0.003	99.62$\pm$0.01
Hyperband	94.16 $\pm$ 0.20	0.967 $\pm$ 0.007	0.979 $\pm$ 0.003	0.974 $\pm$ 0.001	0.963 $\pm$ 0.003	99.41 $\pm$ 0.16
BOHB-Round	94.05 $\pm$ 0.33	0.976$\pm$0.008	0.979 $\pm$ 0.002	0.974 $\pm$ 0.001	0.964 $\pm$ 0.003	99.52 $\pm$ 0.07
BOHB-Bracket	94.24$\pm$0.43	0.972 $\pm$ 0.002	0.979 $\pm$ 0.002	0.975 $\pm$ 0.002	0.965 $\pm$ 0.002	99.56 $\pm$ 0.03

**Figure 1:** Regret on NB301. BOHB-Bracket achieves the lowest median regret; all multi-fidelity methods improve over RS/GS.**Figure 2:** Regret on rbv2_xgboost. BOHB-Round converges fastest; model-guided sampling outperforms pure multi-fidelity exploration.

rbv2_xgboost. BOHB-Round achieves the best mean (0.976 ± 0.008), surpassing Bayesian Optimisation (0.974 ± 0.009) and Hyperband (0.967 ± 0.007). XGBoost’s search space is lower-dimensional and largely continuous, making the RF surrogate easier to fit; model-guided sampling at low fidelity cheaply identifies promising regions before promoting to full training fraction. BOHB-Bracket scores slightly lower (0.972 ± 0.002): the additional refit overhead consumes budget that BOHB-Round can spend on evaluations.

Other benchmarks. On *rbv2_svm* and *rbv2_ranger*, Bayesian Optimisation leads, which is expected given their smaller search spaces where a GP surrogate modelling at full fidelity is highly effective. On *rbv2_glmnet*, Grid Search achieves the best mean, suggesting the parameter space is small enough for uniform coverage to be competitive. LCBench/3945 is dominated by Bayesian Optimisation (99.62 ± 0.01), with BOHB-Bracket a close second (99.56 ± 0.03). Across all six benchmarks, both BOHB variants consistently outperform Random Search and Grid Search, and are competitive with or better than Hyperband.

5 Surrogate Analysis

Figure 3 plots the fraction of configurations proposed by the RF surrogate (versus random sampling) over the course of the run on NB301. Both variants start at zero (no data yet) and climb toward $1 - \rho = 0.67$ as observations accumulate. BOHB-Bracket consistently exceeds BOHB-Round early in the run because it refits with all available data before each bracket, whereas BOHB-Round fits once per round and then freezes the model. The convergence of both curves toward the same plateau confirms that $\rho = 1/3$ random exploration is maintained throughout, as required by Algorithm 2 of Falkner et al.

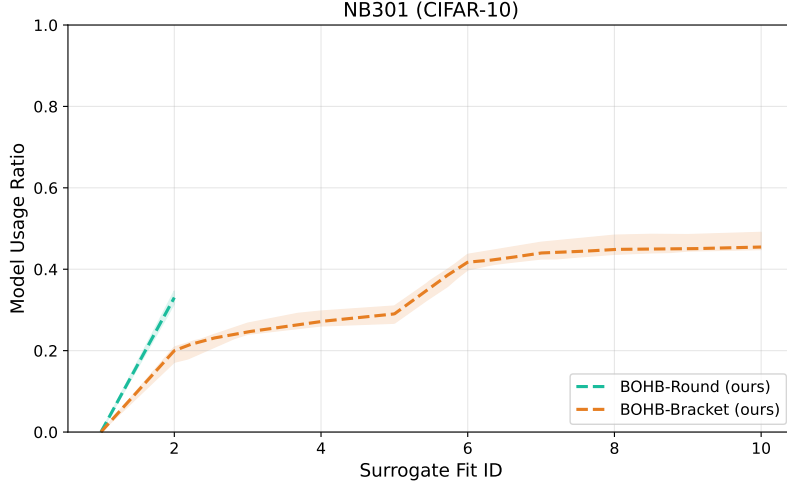


Figure 3: Cumulative model-usage ratio on NB301 (median $\pm$ IQR). BOHB-Bracket reaches a higher model-usage ratio earlier because per-bracket refitting provides fresh data sooner.

6 Conclusions

BOHB matches or exceeds Hyperband on every benchmark tested. On NB301, BOHB-Bracket achieves the highest mean accuracy but with greater variance, reflecting the warmup phase of the surrogate. On `rbv2_xgboost`, BOHB-Round clearly wins, demonstrating the benefit of model-guided sampling on a well-structured continuous space. Per-bracket refitting (BOHB-Bracket) is preferable on high-dimensional or structured benchmarks where earlier brackets provide informative priors; per-round refitting (BOHB-Round) is advantageous when the additional refitting overhead would otherwise consume useful evaluation budget. Label encoding (instead of one-hot) is critical: it keeps NB301’s encoded dimension at 34 rather than 272, reducing $N_{\min}$ from 273 to 35 and allowing the surrogate to activate early enough to influence the run outcome.

Statement of Contributions

Shivam Sen Gupta and Dlyaver Djebarov jointly designed the BOHB implementation. Dlyaver Djebarov implemented the ρ parameter, the per-bracket refitting mode, and the diagnostic logging framework. Shivam Sen Gupta integrated all components, ran the full experiment sweep (420 runs), and wrote the report. AI tools (ChatGPT) were used for language polishing (grammar and wording improvements only).

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